# Prof. Dr. Jonas Karthein

Assistant Professor
Cyclotron Institute / Dep. of Physics & Astronomy
Texas A&M University

karthein@tamu.edu
lab.karthein.com
ORCID 0000-0002-4306-9708
Curriculum vitae, November 2026

## RESEARCH INTERESTS

We study the atomic nucleus as a window into physics beyond the Standard Model, combining precision laser spectroscopy, ion trapping and quantum sensing of short-lived isotopes and radioactive molecules. At the Texas A&M Cyclotron Institute we are building the Universal Action Spectrometer, a trap-based platform for background-free laser spectroscopy with single-ion sensitivity, to measure nuclear charge radii and electroweak, parity-violating nuclear properties such as the anapole moment. The lab leads an inaugural DOE Genesis Mission award on differentiable digital twins and AI-driven autonomous operation of precision experiments, runs collaborative experiments at FRIB, Argonne, Los Alamos and CERN/ISOLDE, and co-leads the NEPTUNE collaboration.

## EDUCATION

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|---|---|
| 2020 | **Doctor rer. nat.**, Physics, Heidelberg University, Germany<br>"Summa Cum Laude"; Advisor: Prof. Dr. Klaus Blaum<br>Thesis: *Next-Generation Mass Spectrometry of Exotic Isotopes and Isomers* |
| 2017 | **M.Sc.**, Physics, Heidelberg University, Germany<br>Thesis: *Precision mass measurements using the Phase-Imaging Ion-Cyclotron-Resonance detection technique* |
| 2015 | **B.Sc.**, Physics, Heidelberg University, Germany |
| 2012 | **Abitur**, Wendalinum Gymnasium, Germany |

## RESEARCH EXPERIENCE

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|---|---|
| 2025 – now | **Assistant Professor**, Texas A&M University, USA |
| 2021 – 2024 | **Humboldt Postdoctoral Fellow**, Massachusetts Institute of Technology (MIT), USA |
| 2017 – 2020 | **CERN Doctoral Fellow**, CERN, Switzerland |
| 2016 – 2017 | **Master's Researcher**, CERN, Switzerland |
| 2012 – 2016 | **Bachelor's Researcher**, Max Planck Institute for Nuclear Physics, Germany |

## GRANTS & FUNDING

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|---|---|
| 2026 – now | **DOE Genesis Mission (inaugural award)** (Lead PI), U.S. Department of Energy, Office of Science, award DE-SC0026653 |
| 2021 – 2025 | **Humboldt Postdoctoral Fellowship** (Fellow), Alexander von Humboldt Foundation, Germany |
| 2017 – 2020 | **CERN Doctoral Scholarship (Gentner Program)** (Fellow), German Ministry for Education and Research |

## AWARDS & HONORS

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|---|---|
| 2025 | MIT Spot Award, MIT School of Science, USA |
| 2023 | FRIB Visiting Scholar Award, Facility for Rare Isotope Beams, USA |
| 2022 | Young Scientist Award, GSI/FAIR (GENCO), Germany |
| 2022 | Dissertation Award, European Physical Society, Spain |
| 2021 | MIT Spot Appreciation Award, MIT School of Science, USA |
| 2021 | Dissertation Award, German Physical Society, Germany |
| 2020 | Dissertation Award, Wilhelm und Else Heraeus Foundation, Germany |
| 2018 | Poster Presentation Award, Winter School on Trapped Charged Particles, France |
| 2017 | Poster Presentation Award, Euroschool on Exotic Beams, France |

## TEACHING

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|---|---|
| 2025 | PHYS 207 - Don't Panic #519-524: Electricity & Magnetism for Scientists and Engineers (97 students) |
| 2025 | PHYS 207 - Don't Panic #507-512: Electricity & Magnetism for Scientists and Engineers (114 students) |
| 2021 | Kaufman Teaching Certificate Program, MIT School of Science, USA |
| 2021 | MIT LEAPS Program Lecturer, MIT, USA |
| 2018 | Teaching Assistant, Heidelberg University, Germany |

## OUTREACH & SERVICE

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|---|---|
| Peer reviewer | Nature (Springer), European Physical Journal A (Springer), Nuclear Physics A (Elsevier), U.S. Department of Energy (DOE SCGSR) |
| 2026 – now | Organizing Committee Leader, 2027 Exotic Beams Summer School, TAMU |
| 2026 | Outreach Lecture for Texas High School Physics Teachers (MIPEP), TAMU |
| 2026 | Outreach Event Demonstrations at the Texas A&M Physics and Engineering Festival, TAMU |
| 2026 | Outreach Lecture to High School Students, Saturday Morning Physics, TAMU |
| 2025 – 2026 | Organizing Committee Member, 2026 Exotic Beams Summer School, Oak Ridge National Lab |
| 2025 | Conference Organizing Committee Member, 2025 Low Energy Community Meeting, TAMU |
| 2025 | Sole Conference Organizer, Workshop on Precision Measurements with Radioisotopes, TAMU |
| 2021 – 2024 | Committee Member, MIT Radiation Protection Committee, USA |
| 2021 – 2024 | Environment, Health & Safety Officer, MIT School of Science, USA |
| 2019 – 2023 | Collaboration Website Design & Maintenance, www.cern.ch/isoltrap |
| 2017 – 2020 | Scientific Outreach, CERN Visitor Program |
| 2014 – 2015 | Conference Organizing Committee Leader, 2015 German Physical Society Spring Meeting |

## MENTORING & SUPERVISION

**Postdoc research advisor**

2. 2025 – now **Brooke Rickey**, TAMU, trap-based single-ion sensitive laser spectroscopy project
1. 2025 – now **Charlotte König**, TAMU & MIT, parity-violation and molecular break-up project

**Graduate research advisor**

7. 2025 – now **Savannah Limarenko**, TAMU, parity-violation and molecular break-up project
6. 2025 – now **Mason Moenter**, TAMU, trap-based single-ion sensitive laser spectroscopy project
5. 2025 – now **Nicholas Schnoor**, TAMU, trap-based single-ion sensitive laser spectroscopy project
4. 2023 – 2024 **Derick Emmanuel Gonzalez-Acevedo**, MIT, radio-frequency trap project
3. 2022 – 2024 **Haruka Kakioka**, MIT, radio-frequency trap and parity violation project
2. 2021 – 2024 **Scott Barrow Moroch**, MIT, parity violation project **Hertz Fellow 2022**
1. 2021 – 2024 **Silviu-Marian Udrescu**, MIT, parity violation project → now Assistant Professor, Princeton University (from January 2027)

**Master's research advisor**

2. 2025 **YeonJun Choi**, TAMU, molecular break-up project
1. 2017 – 2018 **Timo Steinsberger**, CERN, radio-frequency quadrupole trap optimization

**Bachelor's and summer student research advisor**

12. 2026 – now **Huy "Henry" Nguyen**, TAMU, ML simulation emulator
11. 2026 **Meg Gillies (REU)**, TAMU, CID gas cell
10. 2026 – now **Shreya Kulkarni**, TAMU, action spectrometer
9. 2025 – now **Riddhi Chhabra**, TAMU, wavelength meter stabilization
8. 2025 – now **Anna Goodson (TREND)**, TAMU, laser ion source development
7. 2025 – 2026 **Gabe Daniels (TREND)**, TAMU, Cyclotron Yield Chart
6. 2025 **Jack Thomas**, TAMU, 3D printing of electrodes
5. 2021 – 2024 **Andoni Fernandez Chiu**, MIT, hardware and control system
4. 2021 – 2022 **Juan Antonio Luera**, MIT, various sensors and control system
3. 2020 – 2021 **Parashar Mohapatra**, MIT (remote), ion-optical simulations → now Senior Electrical Engineer, Joby Aviation
2. 2020 – 2021 **Sarah Song**, CERN (remote), realistic buffer gas simulations

1. 2019 – 2020 **Adèle Perus**, CERN, ion-optical simulations and lab integration

## THESIS COMMITTEES

2026 – now Ph.D. thesis committee member
2026 – now Master's thesis committee member
2025 Dissertation committee, University of Jyväskylä, Finland

## BEAMTIME PROPOSALS

2025 **FRIB #25015**, oxygen isotopes for nuclear structure (co-author)
2025 **FRIB #25009**, zirconium isotopes for nuclear structure (co-author)
2025 **FRIB #25004**, francium isotopes for nuclear structure (co-author)
2023 **FRIB #23006**, nuclear astrophysics, r-process (spokesperson)
2023 **FRIB #23080**, precision nuclear structure (spokesperson)
2023 **FRIB #23060**, molecules for astrophysics (co-author)
2021 **FRIB #21023**, T-violation (spokesperson)
2021 **FRIB #21015**, precision nuclear structure (co-author)
2021 **INTC-P-612**, precision nuclear structure (co-author)
2017 **INTC-I-182**, Standard Model Tests (spokesperson, letter of intent)

## PUBLICATIONS

26 peer-reviewed articles and preprints, 466 citations (INSPIRE-HEP, November 2026); ORCID 0000-0002-4306-9708

### Pre-Peer-Review

26. CREMA: Collinear Resonance for Exotic Molecules and Atoms
Cruz FCP, Wilkins SG, Brinson AJ, Fischer P, Flayol M, ..., **Karthein J** et al.
arXiv:2610.30522
25. Nuclear charge radii of aluminum isotopes at the proton drip line
Brinson A, Rickey B, Arthuis P, Belley A, Campbell S, ..., **Karthein J** et al.
*Under Review with Nature Physics.* arXiv:2605.09139
** Postdoc Dr. Rickey led the experiment and analysis. The results were published in her dissertation.*
24. Hadronic parity violation: successes, challenges, and future prospects
Gardner S, **Karthein J**, Meißner UG, Muralidhara G, Navratil P, Snow WM.
*Under Review with Elsevier for the Encyclopedia of Nuclear Physics.* arXiv:2605.07826
** Invited review article.*

### Peer-Reviewed

23. Hyperfine structures of 4s $^2S_{1/2}$, 5s $^2S_{1/2}$, 3p $^2P_{1/2,3/2}$, and mixed nd $^2D_{3/2,5/2}$ states in $^{27}$Al I
Rickey BJ, Brinson A, Chakraborty A, Dockery A, Garcia Ruiz RF, **Karthein J** et al.
*Phys. Rev. A 113, 042824 (2026).* doi
** Postdoc Dr. Rickey led the experiment, analysis, and publication.*
22. Precision mass measurement of $^{61}$Zn and updated Ft value for conserved vector current tests
Giesel PF, Atanasov D, Blaum K, Fischer P, **Karthein J** et al.
*Phys. Rev. C 113, 034318 (2026)* doi
21. Commissioning the Resonant Ionization Spectroscopy Experiment at FRIB
Brinson AJ, Rickey BJ, Allmond JM, Dockery A, Chiu AF, ..., **Karthein J** et al.
*Phys. Rev. Res. 8, 013276 (2026)* arXiv:2511.08881 doi
20. Charge radii measurements of exotic tin isotopes in the proximity of N = 50 and N = 82
Gustafsson FP, Rodríguez LV, Garcia Ruiz RF, Miyagi T, Bai SW, ..., **Karthein J** et al.
*Phys. Rev. Lett. 135, 222501 (2025)* arXiv:2504.17060 doi
19. Extending the low-Z "border" of the A=100 region of deformation with precision mass spectrometry of $^{96-98}$Kr
Lunney D, Manea V, Mougeot M, Nies L, Althubiti NA, ..., **Karthein J** et al.
*Phys. Rev. C 112, 034307 (2025)* arXiv:2509.05538 doi
18. Variations in the charge radii of indium isotopes between N = 52 and 82
Vernon AR, Binnersley CL, Sahoo BK, Billowes J, Bissell ML, ..., **Karthein J** et al.
*Phys. Rev. C 111, 064325 (2025)* **Editor's Suggestion** doi

17. Electromagnetic properties of indium isotopes elucidate the doubly magic character of $^{100}$Sn
**Karthein J**, Ricketts CM, Garcia Ruiz RF, Billowes J, Binnersley CL et al.
*Nature Phys. 20, 1719 (2024)* arXiv:2310.15093 doi
16. Electroweak Nuclear Properties from Single Molecular Ions in a Penning Trap
**Karthein J**, Udrescu SM, Moroch SB, Belosevic I, Blaum K et al.
*Phys. Rev. Lett. 133, 033003 (2024)* arXiv:2310.11192 doi
15. Nuclear charge radii of silicon isotopes
König K, Berengut JC, Borschevsky A, Brinson A, Brown BA, ..., **Karthein J** et al.
*Phys. Rev. Lett. 132, 162502 (2024)* arXiv:2309.02037 doi
14. Opportunities for Fundamental Physics Research with Radioactive Molecules
Arrowsmith-Kron G, Athanasakis-Kaklamanakis M, Au M, Ballof J, Berger R, ..., **Karthein J** et al.
*Rep. Prog. Phys. 87, 084301 (2024)* arXiv:2302.02165 doi
13. Isomeric Excitation Energy for 99Inm from Mass Spectrometry Reveals Constant Trend Next to Doubly Magic $^{100}$Sn
Nies L, Atanasov D, Athanasakis-Kaklamanakis M, Au M, Blaum K, ..., **Karthein J** et al.
*Phys. Rev. Lett. 131, 022502 (2023)* arXiv:2306.02033 doi
12. KATRIN background due to surface radioimpurities
Fränkle F, Schaller A, Blaum K, Otten E, Bornschein L, ..., **Karthein J** et al.
*Astropart. Phys. 138, 102686 (2022)* arXiv:2011.05107 doi
11. Mass measurements of $^{99-101}$In challenge ab initio nuclear theory of the nuclide $^{100}$Sn
Mougeot M, Atanasov D, **Karthein J**, Wolf RN, Ascher P et al.
*Nature Phys. 17, 1099 (2021)* arXiv:2109.10673 doi
** MM, DA, and JK share first authorship for equal contributions.*
10. Analysis methods and code for very high-precision mass measurements of unstable isotopes
**Karthein J**, Atanasov D, Blaum K, Lunney D, Manea V, Mougeot M.
*Comp. Phys. Comm. 267, 108070 (2021)* arXiv:2102.10413 doi
9. Masses of short-lived $^{49}$Sc, $^{50}$Sc, $^{70}$As, $^{73}$Br, and stable $^{196}$Hg nuclides
Kulikov I, Algora A, Atanasov D, Ascher P, Blaum K, ..., **Karthein J** et al.
*Nucl. Phys. A 1002, 121990 (2020)* arXiv:2010.15467 doi
8. Examining the N=28 shell closure through high-precision mass measurements of $^{46-48}$Ar
Mougeot M, Atanasov D, Barbieri C, Blaum K, Breitenfeld M, ..., **Karthein J** et al.
*Phys. Rev. C 102, 014301 (2020)* arXiv:2006.02712 doi
7. First glimpse of the N=82 shell closure below Z=50 from masses of neutron-rich cadmium isotopes and isomers
Manea V, **Karthein J**, Atanasov D, Bender M, Blaum K et al.
*Phys. Rev. Lett. 124, 092502 (2020)* arXiv:2001.05075 doi
** VM and JK share first authorship for equal contributions.*
6. Improved stability of multi-reflection time-of-flight mass spectrometers through passive and active voltage stabilization
Wienholtz F, Blaum K, **Karthein J**, Lunney D, Malbrunot-Ettenauer S et al.
*Nucl. Inst. Meth. Phys. B 463, 348 (2020).* doi
5. QEC-value determination for $^{21}$Na$\rightarrow^{21}$Ne and $^{23}$Mg$\rightarrow^{23}$Na mirror-nuclei decays using high-precision mass spectrometry with ISOLTRAP at ISOLDE/CERN
**Karthein J**, Atanasov D, Blaum K, Breitenfeldt M, Bondar V et al.
*Phys. Rev. C 100, 015502 (2019)* arXiv:1906.01538 doi
4. Direct decay-energy measurement as a route to the neutrino mass
**Karthein J**, Atanasov D, Blaum K, Eliseev S, Filianin P et al.
*Hyperfine Int. 240, 61 (2019)* arXiv:1905.05510 doi
3. Long-term monitoring of the internal energy distribution of isolated cluster systems
Breitenfeldt C, Blaum K, George S, Göck J, Guzmán-Ramírez G, **Karthein J**, et al.
*Phys. Rev. Lett. 120, 253001 (2018).* arXiv:1805.11403 doi
2. Photodissociation of an Internally Cold Beam of CH+ ions in a Cryogenic Storage Ring
O'connor AP, Becker A, Blaum K, Breitenfeldt C, George S, ..., **Karthein J**, et al.
*Phys. Rev. Lett. 116, 113002 (2016).* doi
1. The Cryogenic Storage Ring CSR
von Hahn R, Becker A, Berg F, Blaum K, Breitenfeldt C, ..., **Karthein J** et al.
*Rev. Sc. Instr. 87, 063115 (2016)* arXiv:1606.01525 doi

### Theses

2. Next-Generation Mass Spectrometry of Exotic Isotopes and Isomers
   **Karthein J.**
   *Heidelberg University (2020)* doi
1. Precision mass measurements using the Phase-Imaging Ion-Cyclotron-Resonance detection technique
   **Karthein J.**
   *Heidelberg University (2017)* doi

### Other

6. Karthein Lab document templates and style: Typst themes (slides, posters, lecture notes, CV, lab notebook), palette, matplotlib style, guides
   **Karthein J.**
   *GitHub, jonas-ka/lab-templates (2026).* repository
5. Reproducible research talks with Typst + Touying, a curated figure library and Claude Code: pipeline, themes, style, recipe (template)
   **Karthein J.**
   *GitHub, jonas-ka/talks-repo-template (2026).* repository
4. Karthein Lab Website
   **Karthein J.**
   *GitHub, jonas-ka/karthein-lab-website (2026).* repository
3. PI-ICR analysis software
   **Karthein J.**
   *GitHub, Zenodo (2020).* repository
2. Python script to produce publication-ready nuclear chart plots
   **Karthein J.**
   *GitHub, jonas-ka/nuclear-chart-plotter (2019).* repository
1. MRToF analysis software
   **Karthein J.**
   *GitHub (2019).* repository

## PRESENTATIONS

63 talks and posters, 33 invited

63. 2026/12 **Nuclear Seminar, University of Kentucky** **invited**, "Trapped Radioactive Molecules: A New Path for Hadronic Parity Violation Measurements," Lexington, KY, USA *(scheduled)*
62. 2026/11 **TRIUMF Colloquium**, "Agentic Digital Twins and Radioactive Molecules at the Texas A&M Cyclotron Institute," Vancouver, BC, Canada
61. 2026/11 **MUI-QC Workshop** **invited**, "Agentic Digital Twins for Self-Driving Precision Experiments," College Station, TX, USA
60. 2026/10 **Texas A&M University System Artificial Intelligence Research & Innovation Meeting**, "Autonomous Nuclear Experiments: Agentic Digital Twins supported by the DOE Genesis Mission," Kyle Field, Texas A&M University, College Station, TX, USA
59. 2026/09 **AWS**, "Digital Twins for Self-Driving Physics Experiments," virtual
58. 2026/09 **NVIDIA VISION**, "Fast, Differentiable Simulations for Self-Driving Physics Experiments," College Station, TX, USA
57. 2026/08 **RadIs Workshop** **invited**, "Isotope Opportunities at Texas A&M University," Oak Ridge National Laboratory, TN, USA
56. 2026/08 **NAMO Workshop** **invited**, "Agentic Digital Twins for Autonomous Nuclear Precision Facilities," Argonne National Laboratory, IL, USA
55. 2026/08 **AI for Nuclear Summit** **invited**, "Agentic Digital Twins for Autonomous Nuclear Precision Facilities," College Station, TX, USA
54. 2026/08 **Cyclotron Seminar**, "Artificial Intelligence and Machine Learning for Nuclear Physics," College Station, TX, USA
53. 2026/06 **Mitchell Institute Physics Enhancement Program (MIPEP)**, "Introduction to Nuclear Physics," College Station, TX, USA
52. 2026/04 **Ariel Science Workshop 2026** **Invited Panel**, "Experimental horizons for BSM and electroweak interactions using AMO techniques and rare isotope beams," TRIUMF, Vancouver, BC, Canada

51. 2026/03 **Cyclotron Seminar** invited, “The mechanics of money - Personal Finance for Physicists,” College Station, TX, USA
50. 2026/03 **APS Global** invited, “Trapped (Radioactive) Molecules -- A New Path for Laser-based Parity Violation Measurements,” Denver, CO, USA
49. 2026/02 **Saturday Morning Physics Lecture** invited, “Atoms and Molecules as Windows into the Atomic Nucleus,” College Station, TX, USA
48. 2025/10 **Seminar Talk IQSE** invited, “Quantum Sensing of Nuclei,” College Station, TX, USA
47. 2025/08 **ArtSci Quantum Research Meeting**, “Quantum Sensing of Nuclei at NEPTUNE,” College Station, TX, USA
46. 2025/08 **Low-Energy Community Meeting**, “Summary and Findings of Workshop on Precision Measurements of Radioisotopes (Aug. 11-12),” College Station, TX, USA
45. 2025/08 **Workshop on Precision Measurement with Radioisotopes**, “Nuclear Structure Studies at NEPTUNE and Texas A&M Cyclotron Institute,” College Station, TX, USA
44. 2025/01 **Joint TAMU-UT Quantum Workshop**, “Quantum Sensing with Nuclei,” College Station, TX, USA
43. 2024/10 **APS DNP 2024**, “Shedding Light on the Doubly Magic Character of 100Sn (N=Z=50),” Boston, MA, USA
42. 2024/06 **PLATAN24**, “Parity-Violating Nuclear Effects From Single Molecules In A Penning Trap with NEPTUNE,” Jyväskylä, Finland
41. 2024/03 **Texas A&M Physics Colloquium** invited colloquium, “Revealing Unknown Nuclear Properties with Next-Gen Precision Techniques,” College Station, TX, USA
40. 2024/02 **Colorado School of Mines Physics Colloquium** invited colloquium, “Revealing Unknown Nuclear Properties with Next-Gen Precision Techniques,” Golden, CO, USA
39. 2024/02 **LBNL Nuclear Physics Seminar** invited seminar, “Revealing Unknown Nuclear Properties with Next-Gen Precision Techniques,” Berkeley, CA, USA
38. 2023/08 **Argonne Physics Seminar** invited seminar, “Precision Techniques for Pioneering Symmetry and Structure Studies,” Lemont, IL, USA
37. 2023/08 **GAIN23** invited, “Laser Spectroscopy on Radioactive Molecules,” Boston, MA, USA
36. 2023/02 **Winter Workshop on Nuclear Dynamics** invited, “Precision Nuclear Experiments using Ion Traps and Lasers at FRIB,” Puerto Vallarta, Mexico
35. 2023/01 **FRIB Seminar** invited, “Fundamental Symmetries from Precision Experiments using Ion Traps and Lasers,” East Lansing, MI, USA
34. 2022/10 **US Long-Range-Plan Workshop for Fundamental Symmetries** invited, “Symmetry-violating measurements using single molecular ions in a Penning Trap,” East Lansing, USA (virtual)
33. 2022/10 **Department of Energy Institutional Review at MIT**, “Single Trapped Molecular Ions as Quantum Sensors for Symmetry-Violating Nuclear Properties,” Cambridge, MA, USA
32. 2022/10 **EuNPC22** invited symposium, award, “High-precision measurements in the direct vicinity of the doubly magic 100Sn(N=Z=50) at ISOLDE/CERN,” Santiago de Compostella, Spain
31. 2022/09 **Trapped Charged Particles (TCP22)**, “Symmetry-violating nuclear properties from single molecular ions in a Penning trap,” Glashütten, Germany
30. 2022/09 **NuSTAR Conference** invited symposium, award, “High-precision mass measurements in the direct vicinity of the doubly magic 100Sn(N=Z=50) at ISOLDE/CERN,” Darmstadt, Germany
29. 2022/07 **MRTOF Workshop**, “Exotic Molecules and Atoms Lab at MIT,” Darmstadt, Germany
28. 2022/03 **Spring conference of the German Physical Society** invited symposium, “High-precision mass measurements in the direct vicinity of the doubly magic 100Sn(N=Z=50) at ISOLDE/CERN,” virtual
27. 2022/01 **Argonne Physics Seminar** invited seminar, “High-precision mass measurements in the direct vicinity of the doubly magic 100Sn(N=Z=50) at ISOLDE/CERN,” virtual
26. 2022/01 **INT Parity Nonconservation II** invited, “Towards measurements of symmetry-violating nuclear properties using single molecular ions in a Penning trap,” Seattle, WA, USA
25. 2021/11 **Humboldt Networking Conference**, “Parity Violation from Single Molecular Ions in a Penning trap,” virtual
24. 2021/10 **DNP Conference of the American Physical Society** invited, “Towards measurements of symmetry-violating nuclear properties using single molecular ions in a Penning trap,” virtual
23. 2021/08 **Dissertation award symposium** invited symposium, award, “Next-Generation Mass Spectroscopy on Exotic Isotopes and Isomers,” German Physical Society (DPG-SMuK), virtual

22. 2021/08 **Spring conference of the German Physical Society (DPG21)**, “High-precision mass measurements in the direct vicinity of the doubly magic 100Sn(N=Z=50) at ISOLDE/CERN,” virtual
21. 2021/06 **Opportunities with Radioactive Molecules** invited, “Symmetry-violating properties from single molecular ions in a Penning trap,” Massachusetts Institute of Technology, Cambridge, MA, USA
20. 2021/01 **CRIS future workshop**, “Precision Spectroscopy of Single Molecular Ions in a Penning Trap,” virtual
19. 2020/12 **Dissertation award symposium** invited symposium, award, “Next-Generation Mass Spectroscopy on Exotic Isotopes and Isomers,” Heidelberg University, Germany
18. 2020/09 **MIT Laboratory for Nuclear Science Seminar** invited seminar, “High-Precision Mass Measurements of Radionuclides and Their Applications,” Massachusetts Institute of Technology, Cambridge, MA, USA
17. 2019/12 **LBNL Nuclear Physics Forum** invited seminar, “High-Precision Mass Measurements for Nuclear Structure and Astrophysics Studies,” Lawrence Berkeley National Laboratory, Berkeley, CA, USA
16. 2019/09 **MATS workshop**, “ISOLTRAP status,” Oberwesel, Germany
15. 2019/07 **PI-ICR workshop**, “ISOLTRAP’s PI-ICR analysis software,” MPIK, Heidelberg, Germany
14. 2019/06 **Gordon Research Conference** invited, “Precision Mass Measurements of Mirror Nuclei,” Colby-Sawyer College, New London, NH, USA
13. 2019/06 **National Nuclear Physics Summer School**, “Phase-imaging ion-cyclotron-resonance detection with ISOLTRAP at CERN,” Yale University, New Haven, CT, USA
12. 2019/05 **FAIRness19**, “Direct decay-energy measurement as a route to the neutrino mass,” Genova, Italy
11. 2019/03 **ISOLDE seminar** invited seminar, “Probing nuclear isomers using phase-imaging ion-cyclotron-resonance detection with ISOLTRAP at CERN,” CERN, Geneva, Switzerland
10. 2019/02 **Arbeitstreffen Kernphysik** invited, “High-precision mass measurements for nuclear astrophysics studies using ISOLTRAP/CERN,” Schleching, Germany
9. 2018/09 **Trapped Charged Particles (TCP18)**, “Probing nuclear isomers using phase-imaging ion-cyclotron-resonance detection with ISOLTRAP at CERN,” Traverse City, MI, USA
8. 2018/09 **Measurements with an Advanced Trapping System (MATS)**, “Probing nuclear isomers using phase-imaging ion-cyclotron-resonance detection with ISOLTRAP/CERN,” Hermeskeil, Germany
7. 2018/02 **Spring conference of the German Physical Society (DPG18)**, “Phase-imaging ion-cyclotron-resonance detection with ISOLTRAP at CERN,” Bochum, Germany
6. 2018/01 **Winter School on Physics with Trapped Charged Particles** award, “The phase-imaging ion-cyclotron-resonance detection technique used at ISOLTRAP/CERN,” Les Houches, France
5. 2017/12 **ISOLDE workshop**, “The phase-imaging ion-cyclotron-resonance detection technique used at ISOLTRAP/CERN,” CERN, Geneva, Switzerland
4. 2017/08 **Euroschool on Exotic Beams** award, “Implementation of the phase-imaging ion-cyclotron-resonance detection technique at ISOLTRAP/CERN,” Cabourg, France
3. 2017/04 **MPIK Institutional review**, “Precision measurements of nuclear ground-state properties with lasers and traps,” Max Planck Institute for Nuclear Physics, Heidelberg, Germany
2. 2017/03 **Spring conference of the German Physical Society**, “Implementation of the phase-imaging ion-cyclotron-resonance detection technique at ISOLTRAP/CERN,” Mainz, Germany
1. 2016/05 **Solid-state Master’s seminar**, “Order parameter in (un)conventional superconductors,” Heidelberg University, Germany